

Introduction

The mechanical architecture of the proposed semi-humanoid robot constitutes the foundational layer upon which all actuation, sensing, and control strategies are built. It is conceived as a holistic, integrated system comprising four principal physical subsystems, each designed with specific functional objectives yet harmonized to operate as a unified kinematic chain. The overarching design philosophy governing this architecture is "Modularity and Functional Decoupling" – ensuring that each subsystem can be developed, tested, and maintained independently while maintaining seamless interoperability at the system level.

This mechanical design prioritizes three cross-cutting criteria:

Mass Distribution & Inertia Minimization: Heavy actuators are positioned proximally to reduce the dynamic load on distal joints, enhancing both safety and responsiveness.

Structural Rigidity & Impact Resilience: Load-bearing frames are designed using topology-optimized skeletal structures to maximize stiffness-to-weight ratios and withstand shock loads during locomotion.

Application-Specific Adaptability: The system is engineered to accommodate a wide range of tasks through a standardized, rapid-interchange interface at the end-effector level.

The following subsections detail the four core mechanical subsystems, which collectively transform the robot from a collection of inert components into a capable, agile, and practical experimental platform:

1. The Upper Body Subsystem (Manipulation & Perception)

The upper body encompasses the head, neck, torso, shoulders, arms, and hands. It is engineered for high-dexterity interaction, featuring 19-21 active degrees of freedom. The design emphasizes low inertia at the distal ends to enable rapid, safe movements. The neck acts as an active gimbal for visual stabilization, while the torso serves as the central structural hub that houses the compute and power units. The shoulders are configured with high-torque, zero-backlash actuators to ensure precise payload positioning, and the forearms employ remote tendon-driven actuation to reduce mass.

2. End-Effector Plug-and-Play Mechanism (Modular Interchangeability)

A defining feature of this platform is its commitment to rapid reconfiguration. A standardized, quick-change mechanical interface is integrated at the wrist (**end-effector**)

level. This mechanism allows the operator to swap between various tools and grippers (e.g., adaptive hands, suction cups, or specialized sensors) within seconds, without requiring tools or disassembly. This plug-and-play capability, coupled with automatic electrical and communication docking, enables the robot to adapt to diverse tasks—from delicate object manipulation to industrial-grade fastening—making it a truly versatile research and educational testbed.

3. Telescopic Linear Actuator (Vertical Lift & Torso Positioning)

The upper body is mounted atop a telescopic linear actuator that serves as the adjustable "spine" connecting the torso to the mobile base. This actuator provides a controlled vertical lifting range (approx. 200-400 mm travel), enabling the robot to dynamically alter its operational height. This functionality expands the effective workspace, allowing the robot to reach high shelves, interact with objects at floor level, or adjust its center of mass height for improved stability during locomotion or manipulation tasks. The telescopic design ensures a compact stowed height while maintaining robust load-bearing capacity through guided rails and a high-efficiency ball-screw or belt-drive transmission.

4. Mobile Robot Base (Locomotion & Ground Mobility)

Serving as the lower-body locomotion module, the mobile robot base is a differential-drive wheeled platform that provides horizontal mobility across flat and semi-structured indoor environments. It is equipped with a suspension system to absorb terrain irregularities and ensure stable sensor data acquisition during movement. The base houses the main battery pack and low-level motor drivers, and its kinematic structure enables precise positioning through odometry and IMU feedback. This subsystem decouples the complexity of walking gaits (legs) from horizontal translation, ensuring that the upper body remains stable and level during navigation, while still allowing the telescopic actuator to adjust the torso height as needed.

Hand The hand is realized as a versatile under-actuated adaptive gripper rather than a fully actuated dexterous hand.

- A single high-torque actuator drives multiple fingers through a tendon network, reducing weight and complexity.
- Finger joints incorporate passive compliance (flexures or elastomer inserts) so that the fingers conform adaptively to irregular object geometries.

- High-strength synthetic tendons (e.g., Dyneema or Kevlar) with return springs provide both flexion and extension.
- Fingertips are equipped with tactile sensing elements (Force Sensitive Resistors or capacitive pressure arrays) for contact detection and force modulation.

This upper-body architecture provides the kinematic redundancy, low inertia, and adaptive grasping capability required to perform a wide range of craft-related manipulation tasks while remaining compatible with the modular end-effector philosophy of the overall platform.

Upper Body Subsystem

The upper body constitutes the primary manipulation and perception interface of the robot, bridging the gap between the mobile base and the interactive environment. This subsystem is engineered to deliver human-like dexterity, rapid responsiveness, and stable sensory acquisition, all while maintaining a lightweight and structurally efficient design.

The mechanical architecture of the upper body is organized into **three distinct yet interconnected functional units**, arranged in a logical top-down sequence that reflects both the physical assembly and the operational workflow of the system.

(1) The Torso (Central Structural Hub)

The torso serves as the rigid skeletal foundation of the entire upper body. It performs three critical functions:

- **Structural Integration:** It rigidly connects the neck, shoulders, and the telescopic actuator (linking to the mobile base), transmitting dynamic loads without flexural deformation.
- **Component Housing:** It encapsulates the central compute unit ("Brain"), power distribution system ("Heart"), and battery modules within a protective, modular chassis.
- **Thermal Management:** Its topology-optimized, open-frame architecture enables passive convective cooling, utilizing aluminum structural members as distributed heat sinks.

(2) The Neck and Head Assembly (Perception and Stabilization Unit)

This assembly functions as the robot's primary sensory control center and communicative interface. It is engineered as a unified system where the head serves as the sensor-rich payload, and the neck acts as the high-performance active stabilizer that isolates this sensitive payload from the dynamic disturbances of the lower body.

The Head (Sensor Carriage)

The head is designed as an ultra-lightweight housing for the robot's core perception sensors. Its primary design philosophy is **strict mass reduction**. All heavy computational hardware (the "Brain"), power regulation circuits, and cooling fans are offloaded to the torso chassis. The head contains only the sensors themselves and lightweight signal interface boards. This minimizes the moment of inertia at the end of the neck, allowing rapid and energy-efficient movements.

The perception suite integrated into the head includes:

- **Stereo Vision Module:** A wide-baseline stereo camera pair mounted behind a protective, optical-grade visor. This configuration generates high-resolution depth maps (point clouds) via triangulation, enabling the robot to distinguish between static obstacles, dynamic actors (humans), and manipulable objects in real-time.
 - **Microphone Array:** A calibrated array of microphones is integrated into the lateral surfaces of the head shell. By measuring the Time-Difference-of-Arrival (TDOA) of sound waves, the system performs Sound Source Localization (SSL), allowing the robot to "hear" the direction of a speaker and automatically turn its head to face the source.
 - **Thermal and Vibration Isolation:** The camera sensors are thermally decoupled from the outer shell to prevent environmental temperature shifts from affecting calibration or causing lens fogging. Additionally, the camera mountings feature vibration dampeners to isolate the optical path from high-frequency vibrations transmitted through the body during walking.
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The Neck (Active Gimbal)



The neck serves as the critical mechanical interface between the head and the torso. It is not merely a passive support structure; it is an **active, high-speed 3-DOF mechanical gimbal** that performs two essential functions:

1. **Active Stabilization (Decoupling):** The neck actively decouples the orientation of the head from the motion of the torso. During locomotion, the torso naturally exhibits sway, pitch, and vertical oscillations. The neck must counteract these disturbances in real-time to maintain a stable visual horizon for the computer vision algorithms. This is achieved through feedback from Inertial Measurement Units (IMUs) mounted in the head and torso, which allows the neck to compensate for body motion before it affects the cameras.

2. **Orientation and Scanning (Pan, Tilt, and Roll):** The neck provides full orientation control, enabling the robot to:
- **Pan (Yaw):** Scan the environment horizontally to track moving objects, locate sound sources, or survey surroundings.
 - **Tilt (Pitch):** Look downward for terrain inspection or object manipulation (e.g., when the robot needs to see what its hands are doing), and look upward for face-to-face interaction with humans.
 - **Roll:** Tilt the head laterally for expressive, non-verbal communication, making the robot more approachable and natural in social interactions.
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Why a 3-DOF Gimbal Mechanism?



After evaluating various 3-DOF mechanisms for the neck, we selected the **3-DOF Gimbal (Spherical Joint Mechanism)** for the following reasons:

(A) Superior Vibration Isolation:

The gimbal's concentric ring architecture inherently provides mechanical decoupling between axes. Unlike differential or parallel mechanisms, the gimbal allows each axis to be driven independently, with minimal transmission of vibrations from one axis to another.

This is critical because the head's sensors require a stable platform; any high-frequency vibration from the torso or arms that reaches the cameras will degrade depth perception and visual odometry accuracy. The gimbal's design naturally filters out high-frequency disturbances.

(B) Independent Axis Control (Decoupled Kinematics):

In a gimbal mechanism, each degree of freedom (Yaw, Pitch, Roll) is actuated by a separate motor, and the motion of each axis is mechanically independent. This simplifies the control algorithms significantly compared to differential or parallel mechanisms, where motion in one axis requires coordinated movement of multiple motors. For the neck, where fast and precise responses are required for stabilization and gaze control, this decoupled architecture allows for simpler, faster, and more robust control loops.

(C) Low Inertia and High Speed:

Because the head is designed to be ultra-lightweight, and the gimbal's motors are positioned at the base of each ring (not at the end-effector), the effective inertia at the head is minimized. This allows the neck to achieve rapid **saccadic motions** (fast eye-like movements) to redirect focus toward new visual or auditory stimuli within milliseconds, without destabilizing the torso.

(D) Simple Mechanical Implementation with Proven Reliability:

The gimbal mechanism is a well-understood, proven design with extensive use in camera stabilization systems (e.g., drone gimbals). It is mechanically straightforward to fabricate with standard bearings, timing belts, or direct-drive motors, and it offers excellent reliability with minimal backlash when properly engineered.

(E) Human-Like Range of Motion:

The gimbal architecture, when properly designed, can achieve a range of motion that closely matches human neck capabilities (approximately $\pm 90^\circ$ in Yaw, $\pm 60^\circ$ in Pitch, and $\pm 30^\circ$ in Roll). This anthropomorphic range is essential for natural social interaction and effective environmental scanning.

Actuation Strategy for the Neck

Given the priority on speed, responsiveness, and smoothness over raw torque, the neck will employ **direct-drive motors or low-reduction gear systems like planetary gear unit**.

This approach:

- **Minimizes backlash:** Eliminating gear play ensures that the head position is always precisely known and controlled, which is essential for accurate gaze control.

- **Reduces inertia:** Low-ratio gears or direct drive minimizes the reflected inertia at the motor, allowing for rapid acceleration and deceleration.
- **Enables back-drivability:** The neck can be manually moved without damage, which is useful for safety and calibration.

The neck control loop will be tightly integrated with the vision and auditory subsystems, enabling coordinated behaviors such as automatically turning toward detected faces or sound sources, and maintaining eye contact during human-robot interaction.

3. The Arms (Manipulation & Interaction Unit)

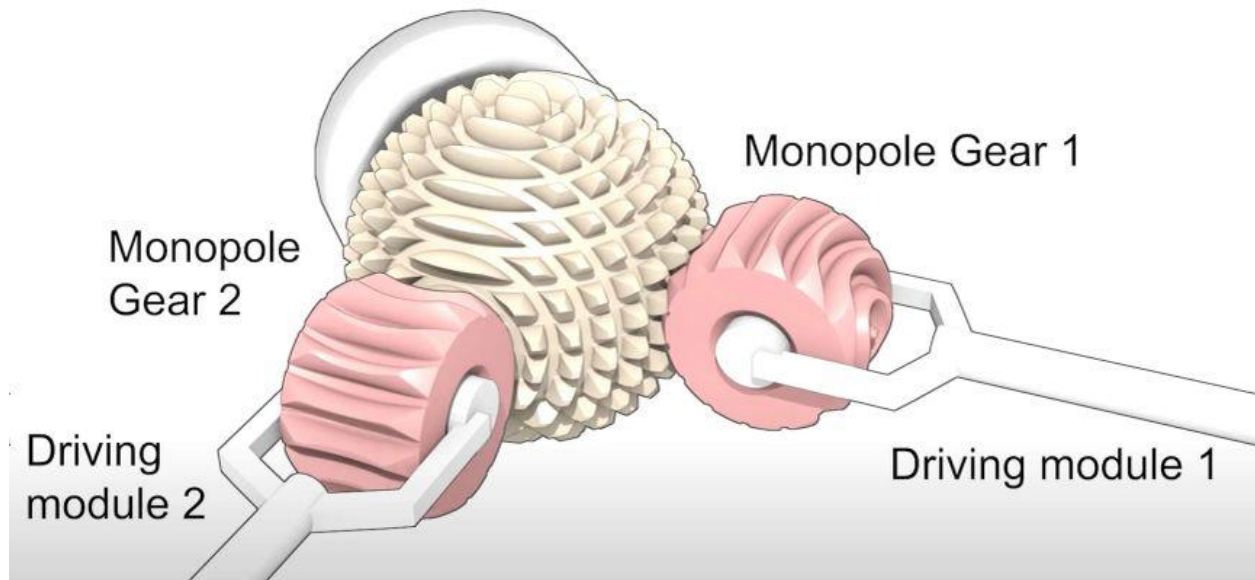
Each arm is a high-dexterity kinematic chain comprising three sequential joint groups, designed under a strict low-inertia philosophy:

- Shoulder (3-DOF)

The shoulder serves as the high-power anchor and primary gross positioning engine for each arm. It bridges the structural core of the torso with the dynamic manipulation system of the arm. Unlike the neck, which prioritizes speed and vibration isolation, the shoulder is engineered for **holding torque, stiffness, and load-bearing capacity**. The critical requirement for this joint is **3-DOF motion** (Pitch, Roll, Yaw) to provide full spatial orientation for the arm, enabling the robot to reach, lift, and manipulate objects across a wide workspace.

To achieve this 3-DOF capability, two candidate mechanisms have been evaluated:

(A) Spherical Gear Mechanism (Spherical Gear Joint)



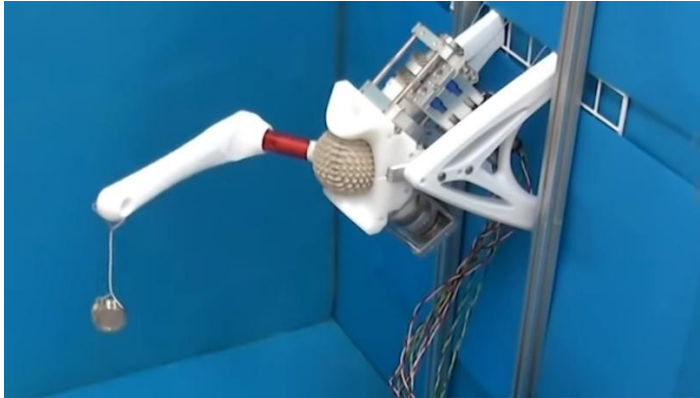
Definition: A Spherical Gear is a specialized mechanical component where the gear teeth are arranged on a spherical surface rather than a flat or cylindrical one. The joint typically consists of a spherical gear (output) that meshes with multiple drive gears (inputs) arranged around it. By controlling the speed and direction of each drive gear, the spherical gear can rotate about multiple axes simultaneously, providing true 3-DOF motion in a compact, single-joint package. This mechanism is conceptually elegant and is still under active research and development in the field of advanced robotics.

How It Works:

- The spherical gear (which is connected to the arm) is surrounded by two or three drive gears positioned orthogonally.
- Each drive gear is actuated by a separate motor.
- By coordinating the rotation of these drive gears, the spherical gear can achieve **Pitch** (tilting up/down), **Roll** (tilting sideways), and **Yaw** (rotating about its vertical axis) – all within a single, compact joint.

Advantages of the Spherical Gear Mechanism:

(1) True 3-DOF in a Single Joint:



Unlike most mechanisms that require multiple joints in series, the spherical gear provides all three rotational degrees of freedom at a single point. This mimics the human ball-and-socket joint more closely and can potentially offer a wider range of motion without mechanical interference between axes.

(2) Simpler Kinematic Control:

Because the joint is essentially a single point of rotation, the forward and inverse kinematics are mathematically simpler compared to mechanisms with offset axes or parallel linkages. The control algorithm can directly command the spherical gear's orientation without needing to compensate for coupled motions.

(3) Compact and Space-Efficient:

The spherical gear mechanism can be housed within a relatively small volume, as all drive components can be clustered around the gear itself. This is beneficial for maintaining a human-like shoulder profile and reducing the overall size of the joint housing.

Disadvantages of the Spherical Gear Mechanism:

(1) Extremely Difficult to Manufacture:

Manufacturing a true spherical gear with precision teeth that mesh uniformly across a 3D surface is **exceedingly challenging**. While 3D printing can produce prototypes, conventional machining (CNC, hobbing, or casting) of spherical gears is prohibitively expensive and requires specialized equipment. This mechanism remains largely in the **research and prototype phase** and is not yet viable for mass production or robust industrial applications.

(2) Limited Load-Bearing Capacity:

Due to the geometry of spherical meshing, the contact area between teeth is often smaller

than in conventional gears, leading to high localized stress and wear. Under heavy payloads or shock loads (common in shoulder applications), the gear teeth are prone to deformation or premature failure.

(3) Sensitivity to Backlash and Manufacturing Tolerances:

Achieving zero-backlash in a spherical gear is extremely difficult due to the complex geometry. Even minor manufacturing inaccuracies or wear over time can introduce significant backlash, which at the end of a long arm translates into unacceptable position errors and vibration.

(4) Complex Assembly and Maintenance:

The assembly of a spherical gear mechanism requires precise alignment of multiple drive gears around the central sphere. Disassembly for maintenance or replacement is time-consuming and requires careful recalibration.

(B) Differential Bevel Gear Drive (Differential Robotic Wrist Mechanism)



Definition: The Differential Bevel Gear Drive is a well-established mechanism that uses two bevel gears (input) meshing with a central bevel gear (output) to generate two degrees of freedom (Pitch and Yaw) through differential motion. In its standard configuration, it provides 2-DOF motion, but it can be modified to include a third axis (Roll) by adding an additional actuator and transmission stage.

How It Works (2-DOF Mode):

- Two independent actuators drive two bevel gears that are positioned at 90° to each other.
- The central output shaft carries the arm or wrist.
- **If both motors rotate in the same direction and speed:** The output shaft rotates about the horizontal axis (Pitch), moving the arm up or down.

- **If both motors rotate in opposite directions and equal speed:** The output shaft rotates about the vertical axis (Yaw), moving the arm left or right.
- By varying the speed ratio between the two motors, the joint can achieve any combination of Pitch and Yaw simultaneously.

How It Works (3-DOF Modified Version):

- To add the third axis (**Roll**), a **third motor** is integrated either:
 - At the base of the joint (behind the differential mechanism) to rotate the entire differential housing.
 - Or at the output shaft (inside the arm) to rotate the end-effector independently.
- With this modification, the mechanism effectively provides 3-DOF: Yaw, Pitch, and Roll.

Advantages of the Differential Bevel Gear Drive:

(1) Proven and Reliable Design:

This mechanism is widely used in robotics, particularly in wrists and shoulders of industrial manipulators. It is a mature, well-understood technology with extensive literature, readily available components, and established manufacturing methods. This reduces development risk and accelerates prototyping.

(2) High Load-Bearing Capacity:

Bevel gears offer large contact surfaces and can be manufactured with hardened steel alloys, providing excellent torque transmission and shock load resistance. This makes them suitable for shoulder applications where payload lifting and holding torque are critical.

(3) Reduced Backlash with Proper Design:

By using precision-ground bevel gears and proper preloading, backlash can be minimized to acceptable levels. While not inherently zero-backlash (like harmonic drives), it can be controlled to a degree that is sufficient for most manipulation tasks.

(4) Scalable and Modular:

The mechanism can be easily scaled up or down, and modular components (gears, bearings, motor mounts) are readily available off-the-shelf. This simplifies sourcing, fabrication, and maintenance.

(5) Cost-Effective Manufacturing:

Bevel gears are standard mechanical components that can be produced using conventional CNC machining, casting, or even 3D printing for prototyping. Compared to spherical gears, the manufacturing cost is significantly lower and more predictable.

Disadvantages of the Differential Bevel Gear Drive:

(1) Requires a Third Axis Modification:

In its base 2-DOF configuration, the mechanism lacks Roll. Adding the third axis requires an additional motor and transmission, which increases the complexity, weight, and volume of the joint.

(2) Coupled Kinematics:

Unlike the spherical gear, the Pitch and Yaw motions are not independent; they are coupled. The control system must use differential speed calculations (adding and subtracting motor speeds) to generate the desired output motion. This requires more complex software algorithms.

(3) Limited Range of Motion (Potential Interference):

The range of motion in Pitch and Yaw can be restricted by the physical arrangement of the bevel gears and the housing, especially when aiming for a full hemisphere of motion like the human shoulder.

(4) Lower Inherent Stiffness in Roll:

The Roll axis, being an additional stage, may introduce additional compliance or backlash if not properly engineered. This requires careful bearing selection and shaft design.

(3) The Elbow Joint

Functional Objective:

The elbow serves as a robust hinge joint connecting the upper arm (humerus) to the forearm. It provides a single degree of freedom (1-DOF) for **flexion and extension** (bending and straightening the arm). While its kinematic function is mechanically straightforward compared to the shoulder or wrist, its design is critical for two reasons: it bears a significant portion of the payload moment (especially when the arm is extended horizontally), and it must maintain a low profile to preserve the arm's anthropomorphic proportions. The primary design goal for the elbow is to achieve **high torque density, low**

backlash, and minimal distal mass, ensuring that the arm remains responsive and energy-efficient during dynamic tasks.

Kinematic Architecture:

- **Type:** 1-DOF Hinge Joint
 - **Motion:** Pure rotational motion about a single fixed axis, providing flexion (reducing the angle between the upper arm and forearm) and extension (increasing the angle).
 - **Range of Motion:** Approximately 0° to 140° , allowing the arm to fold fully for compact stowage and to reach full extension for maximum workspace reach.
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Mechanical Design Considerations (The Challenge):

The elbow is a **cantilevered load-bearing joint**. When the arm is fully extended horizontally (e.g., reaching forward to pick up an object), the elbow experiences a significant bending moment due to the weight of the forearm, wrist, hand, and payload. If a motor were mounted directly at the joint axis (direct-drive), it would need to be impractically large and heavy to generate the required holding torque.

Solution: Belt and Pulley Transmission

To address this, we employ a **belt and pulley transmission system** between the motor and the joint axis. The motor is **mounted proximally** – meaning it is positioned closer to the shoulder (inside the upper arm), not at the elbow joint itself. This approach has several critical benefits:

(1) Torque Amplification:

By using a small pulley on the motor shaft and a larger pulley on the elbow output shaft (a gear reduction ratio of approximately 3:1 to 6:1), the motor's torque is mechanically multiplied. This allows a relatively small, lightweight motor to generate the high holding torque required to resist the bending moment of the extended arm.

(2) Optimal Mass Distribution (Low Inertia):

Placing the motor's mass inside the upper arm (proximal) rather than at the elbow (distal) significantly reduces the moment of inertia of the forearm. This is crucial for dynamic performance, allowing the arm to accelerate and decelerate rapidly without overshooting and without requiring excessive torque from the shoulder motor. The forearm remains lightweight and agile.

(3) Impact Protection:

The belt acts as a mechanical fuse or damper. Unlike rigid gear trains, the belt can absorb shock loads and high-frequency vibrations, protecting the motor from damage during accidental impacts or hard landings. This increases the reliability and lifespan of the joint.

(4) Reduced Backlash and Noise:

Modern toothed belts (e.g., HTD or GT series) offer minimal backlash and operate quietly with proper tensioning. This ensures precise and smooth motion without the audible grinding or play that is common in low-cost spur gears.

(4) The Wrist Joint (2-DOF Orientation Interface)

Functional Objective:

The wrist serves as the critical mechanical interface between the forearm and the end-effector (hand, gripper, or specialized tool). Its primary function is to provide **precise orientation control** of the end-effector relative to the arm, enabling the robot to align its tools with objects in the environment. Unlike the shoulder, which handles gross positioning and heavy payloads, the wrist is engineered for **precision, low inertia, and compactness**. It is required to provide **2-DOF motion**:

1. **Pitch (Flexion/Extension):** Tilting the wrist forward and backward (up and down) to orient the hand for grasping objects from above or below.
2. **Roll (Supination/Pronation):** Rotating the wrist about its longitudinal axis to turn the hand (e.g., for turning a doorknob, pouring liquid, or using a screwdriver).

The wrist must achieve these motions with high precision, minimal backlash, and rapid response, all while maintaining a compact form factor and low mass to avoid increasing the inertia of the distal arm.

Kinematic Architecture:

- **Type:** 2-DOF Differential Bevel Gear Mechanism (same principle as the shoulder, but without modification for a third axis)
- **Motion:**
 - **Pitch (Flexion/Extension):** Achieved when both bevel gears rotate in the **same direction and at the same speed**.

- **Roll (Supination/Pronation):** Achieved when both bevel gears rotate in **opposite directions** (one clockwise, the other counterclockwise) at the same speed.

Mechanism Description: The Differential Bevel Gear Drive



The wrist mechanism is a **standard 2-DOF Differential Bevel Gear Drive**, identical in principle to the shoulder's base configuration but without the added complexity of a third axis for Yaw. It consists of the following components:

(1) Two Input Bevel Gears (Driving Gears):

- These are two identical bevel gears positioned at 90° to each other.
- Each input gear is independently driven by a dedicated actuator (motor).
- The motors are typically mounted **proximally** (inside the forearm) to keep the wrist itself lightweight and compact.
- Power is transmitted from the motors to the input gears via **bevel gear shafts, timing belts, or a combination of both.**

(2) One Output Bevel Gear (Driven Gear):

- This gear is mounted on the output shaft that connects directly to the end-effector.
- It meshes with both input gears simultaneously.
- The output shaft is supported by high-precision bearings (e.g., angular contact bearings) to handle both radial and axial loads.

(3) The Differential Motion Principle:

Motor 1 (Left Input Gear)	Motor 2 (Right Input Gear)	Resulting Wrist Motion
Rotates Clockwise (CW) at Speed X	Rotates Clockwise (CW) at Speed X	Pitch (Flexion): Wrist rotates downward (flexing forward).
Rotates Counterclockwise (CCW) at Speed X	Rotates Counterclockwise (CCW) at Speed X	Pitch (Extension): Wrist rotates upward (extending backward).
Rotates Clockwise (CW) at Speed X	Rotates Counterclockwise (CCW) at Speed X	Roll (Supination): Wrist rotates about its longitudinal axis in one direction (e.g., palm up).
Rotates Counterclockwise (CCW) at Speed X	Rotates Clockwise (CW) at Speed X	Roll (Pronation): Wrist rotates about its longitudinal axis in the opposite direction (e.g., palm down).

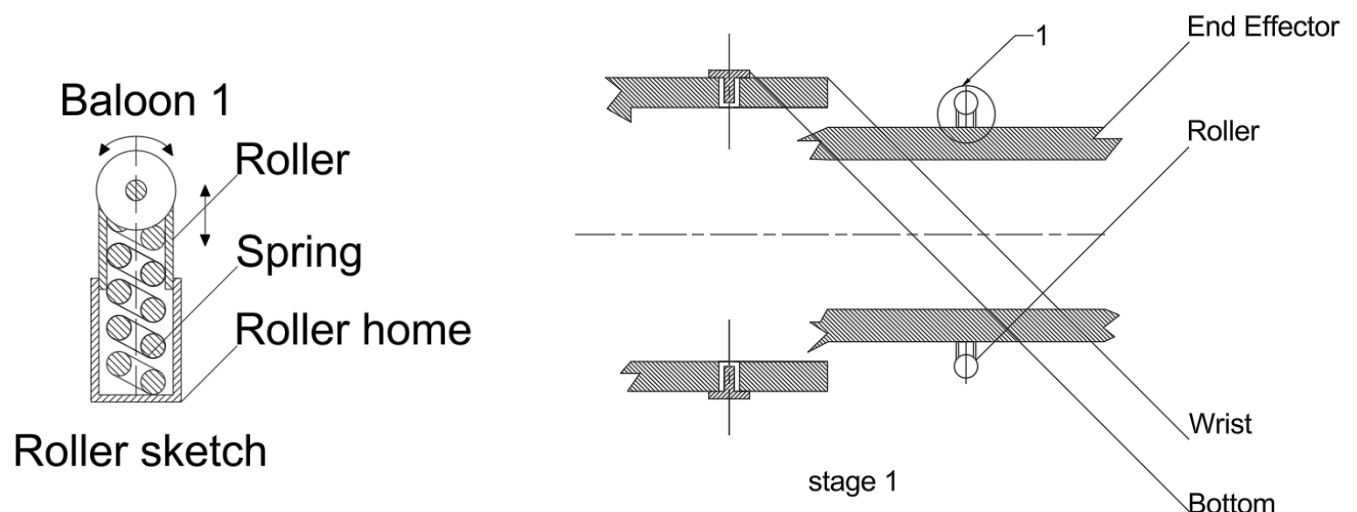
Modular Plug & Play Mechanism: Wrist-to-End-Effector Interface

To achieve the primary goal of modularity, the system is designed to allow a single robotic wrist to seamlessly accommodate multiple interchangeable end effectors for diverse applications. To ensure rapid and tool-less changeovers, a standardized plug-and-play mechanical interface is implemented. All end effectors share an identical mounting base, while the wrist assembly is equipped with a mating receptacle designed for instant locking and unlocking.

The mechanism utilizes a spring-loaded roller locking system. The end effector's base features a distributed set of slots (recesses) on its sides. Correspondingly, the wrist receptacle houses an equal number of spring-loaded rollers, with a dedicated slot recessed into its inner surface to capture each roller.

Component Breakdown (Base on Stage 1 Setup):

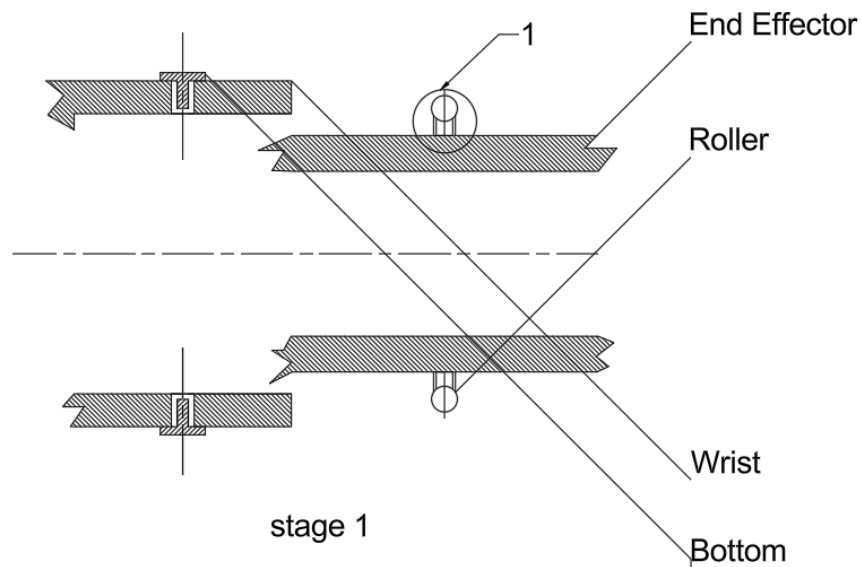
The primary components illustrated in the initial setup include the End Effector (the interchangeable tool), the Wrist (the receiver base), the Roller (the locking actuator), the internal Spring (providing upward biasing force), and the Roller Home (the vertical housing cavity). In Stage 1, the end effector is positioned above the wrist and has not yet initiated engagement.



Operational Stages:

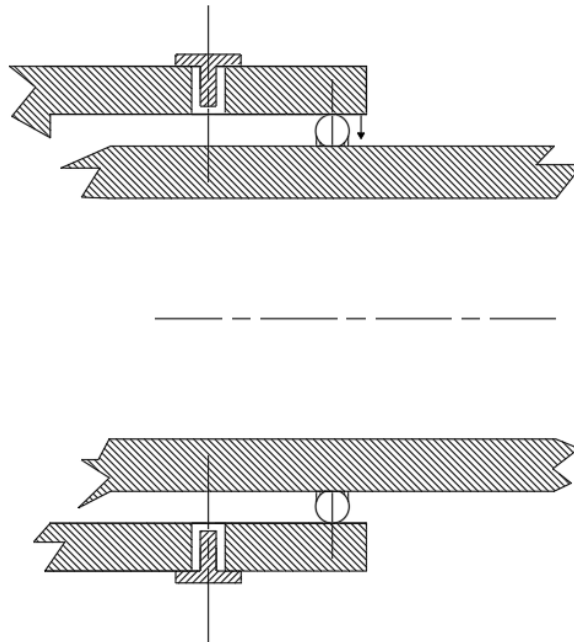
The locking sequence progresses through three distinct mechanical stages:

- Stage 1:



Initial Positioning: The end effector is positioned above the wrist, remaining distant and stationary with no physical contact between their respective bases. The rollers are idle within their housings, held in their resting position by the springs

- **Stage 2:**

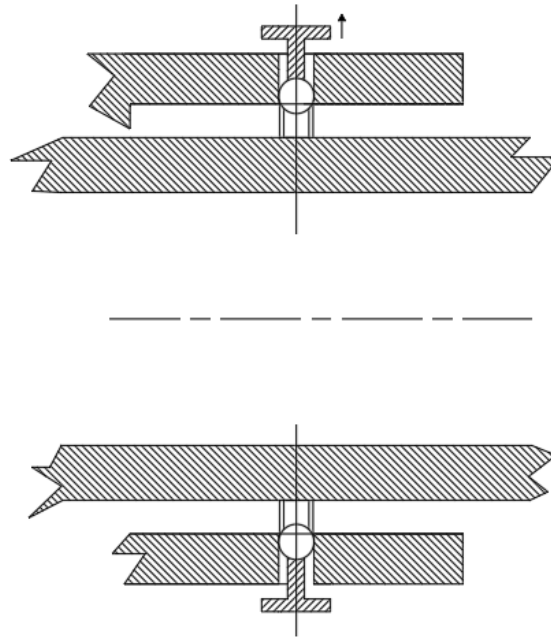


stage 2

Insertion

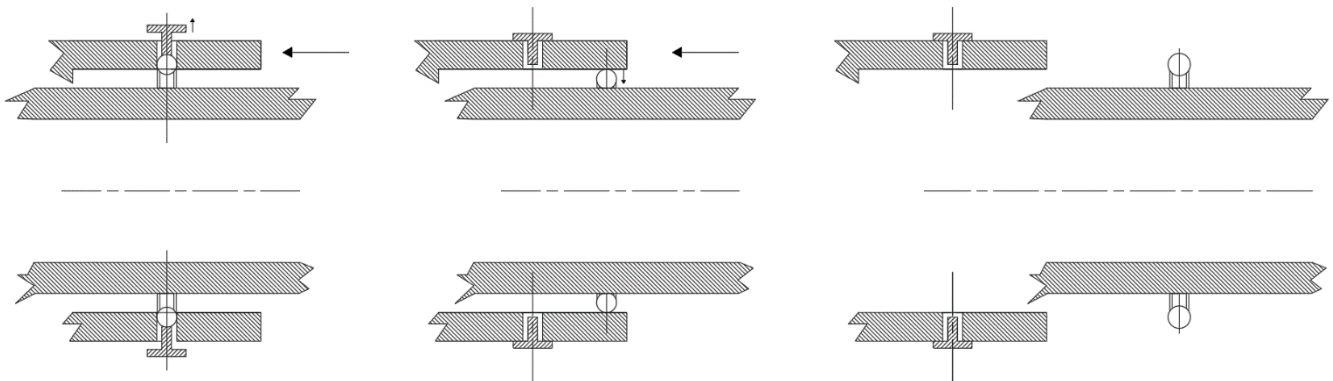
and Spring Compression: As the end effector moves downward, its base contacts the rollers. The downward force pushes the rollers against their springs, compressing them into their homes. The end effector continues to slide downward over the rollers until it reaches the designated alignment depth.

Stage 3:



stage 3

Locking and Snap-In Engagement: Once the end effector's slots align perfectly with the rollers' positions, the compressed springs instantly release their stored energy. This forces the rollers to snap upward and securely lodge into the end effector's slots. This engagement creates a rigid mechanical lock, rendering the end effector irremovable from the wrist. The top section of the rollers now protrudes above the surface, functioning physically as release buttons.



- **Release and Interchangeability:**

To reverse the process and change the tool, the operator applies downward pressure to the protruding roller buttons. This action compresses the springs and retracts the rollers back into their homes, clearing the mechanical blockage. The end effector can then be effortlessly pulled upwards and detached, ready for a different tool to be inserted following the same three-stage sequence.

Telescopic Linear Actuator



The vertical mobility of the upper body is achieved through a multi-stage electric telescopic linear actuator. This solution was deliberately selected over hydraulic or pneumatic alternatives to eliminate the additional mass, complexity, and maintenance requirements associated with oil pumps, reservoirs, or air compressors.

The actuator is driven by an electric motor coupled to a power-screw mechanism (lead screw or ball screw). The nested tubular stages extend and retract in a controlled manner, providing a continuous height adjustment range. The minimum height from the top surface

of the mobile base to the base of the upper body is 40 cm, while the maximum extended height reaches 150 cm, resulting in an effective stroke of approximately 110 cm.

To ensure high rigidity and resistance to bending moments and lateral loads generated by upper-body movements, the telescopic column incorporates linear guides (rails) and precision bearings. These guiding elements maintain precise alignment and significantly reduce deflection under dynamic operating conditions.

A key advantage of the power-screw approach, particularly when using a lead screw, is its inherent self-locking characteristic. This property allows the column to remain securely in position even when electrical power is removed, providing an important passive safety feature.

Proper cable and signal management is integrated into the design. Internal or externally guided routing paths are provided for power and communication cables traveling to the upper body, preventing fatigue, entanglement, or damage during repeated extension and retraction cycles.

Position feedback is obtained through an encoder (absolute or incremental) mounted on the motor or on one of the moving stages, enabling the control system to determine the exact height of the upper body in real time.

Additional safety features include mechanical end stops at both extremes of travel, limit switches that cut power at the maximum and minimum heights, and the optional integration of an electromagnetic brake for enhanced holding capability when required.

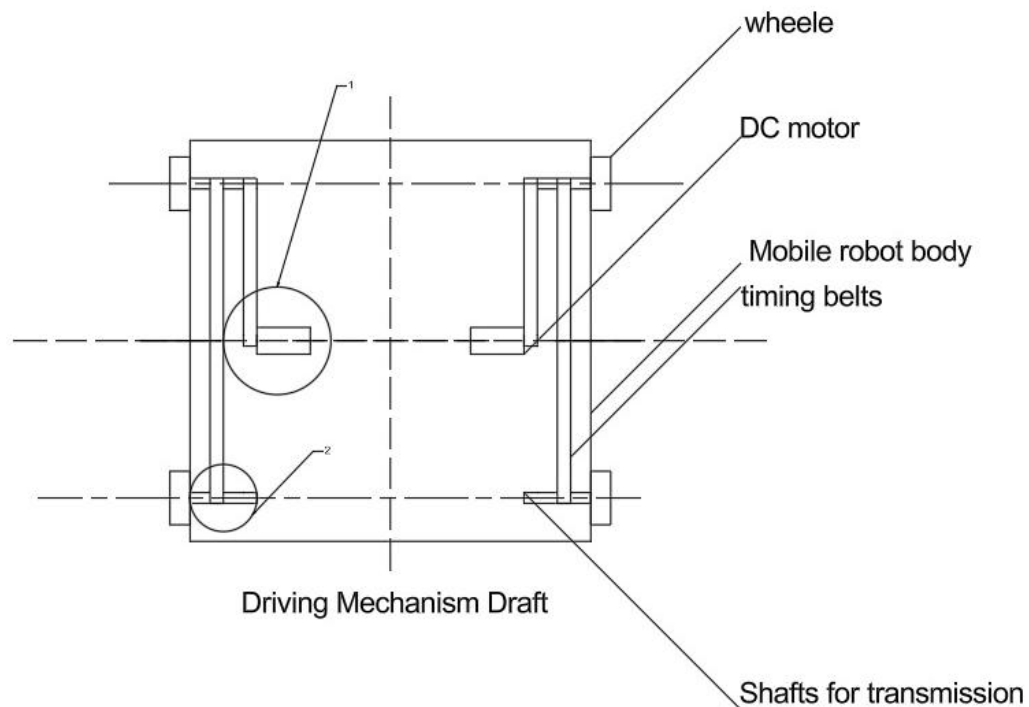
From a structural perspective, the fully extended column is subjected to significant bending moments caused by the mass of the upper body and the dynamic forces generated by arm motion. The combination of nested-stage construction and linear guides is therefore essential to resist buckling and maintain acceptable stiffness throughout the working range.

The final number of telescopic stages, the specific type of power screw (lead screw versus ball screw), and the detailed mounting interfaces will be determined during the detailed design phase once the total payload and required stiffness have been fully quantified.

This electric multi-stage telescopic system enables the robot to adapt its working height to different tasks while allowing the complete platform to collapse into a compact configuration for storage and transportation.

Four-Wheeled Differential-Drive Mobile Base

The locomotion system of the platform is realized as a four-wheeled mobile base employing a **differential-drive configuration**. The system consists of two independently actuated drive sides, with one DC motor dedicated to the right-side wheels and another DC motor dedicated to the left-side wheels. Each motor transmits mechanical power to the two wheels located on its respective side through a timing-belt and pulley transmission system.



Differential-Drive Principle

Differential drive generates motion by independently controlling the rotational speed and direction of the left and right wheel pairs. When both sides rotate at the same speed and in the same direction, the robot moves approximately straight forward or backward. When the two sides rotate at different speeds, the robot follows a curved trajectory. By driving the two sides in opposite directions, the robot can rotate about a point near the center of the mobile base, enabling in-place turning.

The basic relationship between the left and right wheel velocities determines both the translational and rotational motion of the platform. This provides the robot with sufficient maneuverability for navigating indoor workspaces and positioning its upper body relative to workstations and objects.

Mechanical Architecture

The mobile base consists of a rigid rectangular chassis that supports the four wheels, drive motors, transmission components, and the telescopic column assembly. The four conventional wheels are arranged in two mechanically connected pairs:

- **Right-side wheel pair:** driven by the right DC motor.
- **Left-side wheel pair:** driven by the left DC motor.

The wheels on each side are mechanically synchronized through a timing-belt transmission. The drive system therefore requires only **two primary drive motors**, while still providing powered motion to all four wheels.

The chassis also provides the main structural interface between the mobile base and the telescopic column. Its dimensions and wheel spacing will be determined during the detailed design stage based on the total robot mass, center-of-mass location, required stability, and available workspace.

Motor and Timing-Belt Transmission

Each side of the mobile base is actuated by a dedicated DC motor mounted to the chassis through a rigid motor bracket. The motor shaft carries a **driving pulley**, which transmits torque through a timing belt to the corresponding driven pulley connected to the wheel-drive shaft.

The general transmission arrangement is:

DC Motor → Driving Pulley → Timing Belt → Driven Pulley → Drive Shaft → Two Wheels

A timing-belt system is selected to provide positive, synchronized power transmission between the motor and the driven shaft while maintaining a relatively compact mechanical arrangement. The pulley ratio can also be selected during the detailed design stage to achieve the required compromise between wheel torque and vehicle speed.

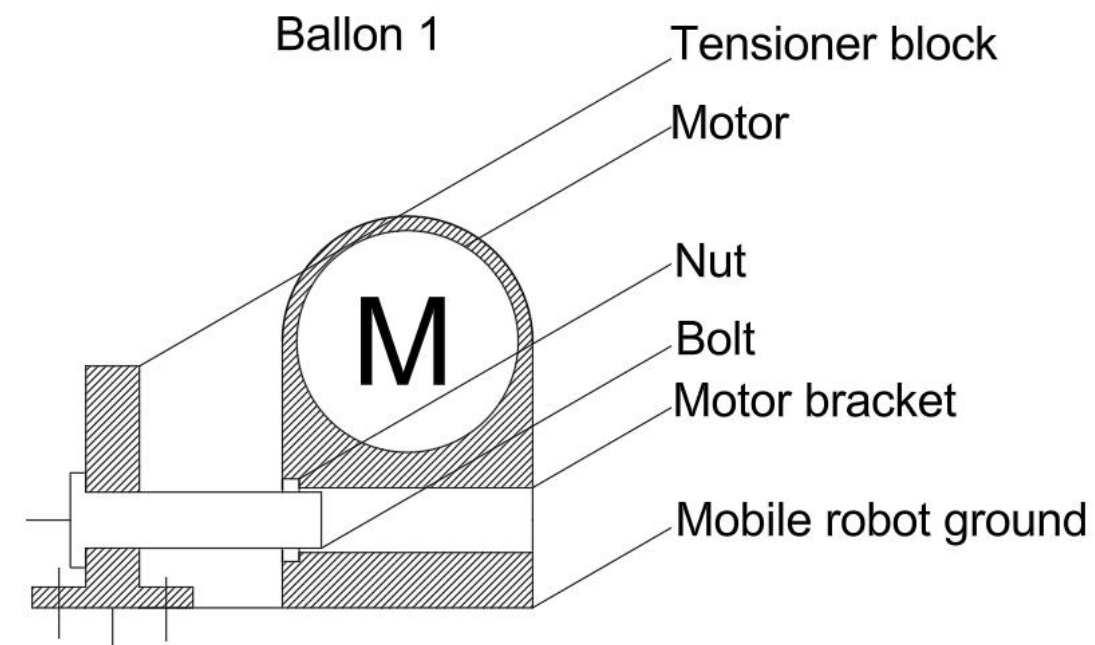
The drive shafts are supported by appropriate bearings mounted to the mobile-base structure to maintain alignment and withstand the radial and torsional loads generated during operation.

Belt Tensioning Mechanism

A dedicated **belt-tensioning mechanism** is incorporated into each timing-belt transmission to maintain the required belt tension during operation. As illustrated in the preliminary design, the tensioner consists of an adjustable tensioner block with a pulley or idler element that applies controlled tension to the belt.

The tensioner assembly is mounted to the chassis through an adjustable mechanism, allowing the belt tension to be fine-tuned during assembly and maintenance. The tensioning system helps maintain proper engagement between the belt and pulleys, reducing the possibility of belt slack, tooth skipping, vibration, and transmission losses.

The final tensioner geometry and adjustment range will be determined according to the selected timing belt, pulley dimensions, center distance, and required transmission torque.



draft for belt tensioning idea

Design Considerations

The detailed mechanical design of the mobile base will determine the final:

- Wheel diameter and width
- Wheelbase and track width
- Motor specifications

- Pulley sizes and transmission ratio
- Timing-belt type and length
- Drive-shaft diameter
- Bearing arrangement
- Chassis dimensions and material
- Belt-tensioning mechanism

These parameters will be selected based on the **total robot mass, required travelling speed, acceleration, available motor torque, floor conditions, and stability requirements**, particularly when the telescopic column is extended and the upper body is performing manipulation tasks.